

UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO

FACULTAD DE CIENCIAS, 2027-I

FUNDAMENTOS DE BASES DE DATOS



PRÁCTICA 01: Bitácora de Saúl Vega Navas

PROFESOR:

Gerardo Avilés Rosas

AYUDANTES DE TEORÍA:

Oscar Daniel Flores Linares

Claudia Itzel Gutiérrez Sánchez

AYUDANTES DE LABORATORIO:

Ricardo Badillo Macías

Rocío Aylin Huerta González

Bitácora

Sistema operativo y versión.

- **Memoria:** 7.6 GB
- **Procesador:** 4 x Intel® Core™ CPU @ 2.60GHz
- **Sistema Operativo:** Linux
- **Entorno:** GNOME 50.4
- **Kernel Version:** Linux 6.18.45-1-MANJARO
- **Arquitectura:** x86_64

Distribución.

- Manjaro.

Versión de la instalación.

```
PostgreSQL 18.6 (Debian 18.6-1.pgdg13+2) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
```

```
> docker run -it --link postgres:postgres postgres psql -h postgres -U postgres
WARNING: Links on the default bridge network are deprecated and will be removed in a future release. Use a custom network instead.
Password for user postgres:
psql (18.6 (Debian 18.6-1.pgdg13+2))
Type "help" for help.

postgres=# SELECT version();
              version
-----
PostgreSQL 18.6 (Debian 18.6-1.pgdg13+2) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
(1 row)

postgres=#
```

Tiempo requerido.

- 65 minutos.

Explicación del paso a paso que realizaste con sus respectivas capturas de pantallas.

1. Instalación de Docker y Docker Compose.

```
sudo pacman -Syu docker docker-compose
```

```
navega@navega:~$ sudo pacman -Syu docker docker-compose
:: Synchronizing package databases...
core is up to date
extra is up to date
multilib is up to date
error: failed retrieving file 'core.db' from mirrors2.manjaro.org : Resolving timed out after 10001 milliseconds
error: failed retrieving file 'extra.db' from mirrors2.manjaro.org : Resolving timed out after 10001 milliseconds
warning: docker-1:29.7.2-1 is up to date -- reinstalling
:: Starting full system upgrade...
resolving dependencies...
looking for conflicting packages...

Packages (3) thunderbird-154.0-1  docker-1:29.7.2-1  docker-compose-5.5.0-1
Total Download Size:    91.16 MiB
Total Installed Size:  456.93 MiB
Net Upgrade Size:       45.24 MiB

:: Proceed with installation? [Y/n] y
:: Retrieving packages...
docker-compose-5.5.0-1-x86_64             11.0 MiB 1898 KiB/s 00:06 [#####] 100%
thunderbird-154.0-1-x86_64                80.2 MiB  2.57 MiB/s 00:31 [#####] 100%
Total (2/2):                               91.2 MiB  2.92 MiB/s 00:31 [#####] 100%
(3/3) checking keys in keyring
(3/3) checking package integrity
(3/3) loading package files
(3/3) checking for file conflicts
(3/3) checking available disk space
:: Running pre-transaction hooks...
(1/2) Creating Timeshift snapshot before upgrade...
==> skipping timeshift-autosnap due skipRsyncAutosnap in /etc/timeshift-autosnap.conf set to TRUE.
(2/2) Remove Thunderbird Distribution Settings
:: Processing package changes...
(1/3) reinstalling docker
(2/3) installing docker-compose
(3/3) upgrading thunderbird
:: Running post-transaction hooks...
(1/7) Creating system user accounts...
(2/7) Reloading system manager configuration...
(3/7) Enqueuing marked services...
(4/7) Arming ConditionNeedsUpdate...
(5/7) Updating icon theme caches...
(6/7) Copy Thunderbird Distribution Settings
(7/7) Updating the desktop file MIME type cache...
$ docker --version
Docker version 29.7.2, build a7dca6fdb
```

II. Habilitación del servicio de Docker y configuración del acceso personal.

```
sudo systemctl enable --now docker
sudo usermod -aG docker $USER
newgrp docker
```

```
navega@navega:~$ docker run hello-world
failed to connect to the docker API at unix:///var/run/docker.sock; check if the path is correct and if the daemon is running: dial unix /var/run/docker.sock: connect: no such file or direct
ory
$ sudo systemctl enable --now docker
[sudo] password for navega:
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service' - '/usr/lib/systemd/system/docker.service'.
$ sudo usermod -aG docker $USER
$ newgrp docker
```

III. Verificación de servicio activo.

```
docker run hello-world
```

```
$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f5586f7dd8: Pull complete
4ee71e42bf5: Download complete
Digest: sha256:5d0d36e25913fc30f98b972bd359cc2cddb48098cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

IV. Descarga de la imagen de PostgreSQL.

```
docker pull postgres
```

```

$ docker pull postgres
Using default tag: latest
latest: Pulling from library/postgres
6310eb1bf42: Pull complete
af2a6480a58: Pull complete
5ee6dca7f69: Pull complete
ace4b5b9a91: Pull complete
1bb54e87727: Pull complete
e4633aeec9b5: Pull complete
c5694bf1357: Pull complete
27c0e09f5d7d: Pull complete
8d8422da297d: Pull complete
6463cb5c763: Pull complete
ae0ab777584: Pull complete
5f29c52d44e: Pull complete
4ee663ab8f5: Pull complete
b41cb76c0e1d: Download complete
6f168415223f: Download complete
Digest: sha256:4ef4dbc939d61acea5771265ddb4b4ab27419c913f94cca8cd57cb3ea3c2280
Status: Downloaded newer image for postgres:latest
docker.io/library/postgres:latest
$ docker run -d \
--name postgres \
-e POSTGRES_PASSWORD=mysecretpassword \
-p 127.0.0.1:5432:5432 \
-v postgres_data:/var/lib/postgresql \
postgres
a9c88db2b36845ee45e687f5d4497e95ba26db122234a1325ce8c765aeb3b1
$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
a9c88db2b368   postgres  "docker-entrypoint.s..." 2 minutes ago  Up 2 minutes  127.0.0.1:5432->5432/tcp            postgres
$ docker run -it --link postgres:postgres postgres psql -h postgres -U postgres
psql (18.6 (Debian 18.6-1.pgdg13+2))
Type "help" for help.

postgres=#

```

v. Inicialización de conexión con el contenedor.

```
docker run -it --link postgres:postgres postgres psql -h postgres -U postgres
```

```

CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
a9c88db2b368   postgres  "docker-entrypoint.s..." 2 minutes ago  Up 2 minutes  127.0.0.1:5432->5432/tcp            postgres
$ docker run -it --link postgres:postgres postgres psql -h postgres -U postgres
WARNING: Links on the default bridge network are deprecated and will be removed in a future release. Use a custom network instead.
Password for user postgres:
psql: error: connection to server at "postgres" (127.17.0.2), port 5432 failed: FATAL: password authentication failed for user "postgres"
$ docker run -it --link postgres:postgres postgres psql -h postgres -U postgres
WARNING: Links on the default bridge network are deprecated and will be removed in a future release. Use a custom network instead.
Password for user postgres:
psql (18.6 (Debian 18.6-1.pgdg13+2))
Type "help" for help.

postgres=#

```

VI. Descarga de DBeaver.

```
sudo pacman -S dbeaver
```

```

$ sudo pacman -S dbeaver
[sudo] password for navega:
resolving dependencies...
:: There are 6 providers available for java-runtime=21:
   Repository extra
    1) jdk-openjdk  2) jdk21-openjdk  3) jdk25-openjdk  4) jre-openjdk  5) jre21-openjdk  6) jre25-openjdk

Enter a number (default=1): 1
looking for conflicting packages...

Packages (4) java-environment-common-3-6  java-runtime-common-3-6  jdk-openjdk-26.0.2.1.u1-1  dbeaver-26.1.5-1
Total Download Size: 497.36 MiB
Total Installed Size: 711.35 MiB

:: Proceed with installation? [Y/n] y
:: Retrieving packages...
java-environment-common-3-6-any                2.7 KiB  540  B/s 00:05 [#####] 100%
java-runtime-common-3-6-any                    3.0 KiB   841  B/s 00:06 [#####] 100%
dbeaver-26.1.5-1-x86_64                       90.2 MiB 1522 KiB/s 01:06 [#####] 100%
jdk-openjdk-26.0.2.1.u1-1-x86_64              323.9 MiB 2.93 MiB/s 01:50 [#####] 100%
Total (4/4)                                   497.4 MiB 2.89 MiB/s 02:52 [#####] 100%
error: failed retrieving file 'jdk-openjdk-26.0.2.1.u1-1-x86_64.pkg.tar.zst' from mirrors2.manjaro.org : Operation too slow, Less than 1 bytes/sec transferred the last 10 seconds
(4/4) checking keys in keyring [#####] 100%
(4/4) checking package integrity [#####] 100%
(4/4) loading package files [#####] 100%
(4/4) checking for file conflicts [#####] 100%
(4/4) checking available disk space [#####] 100%
:: Processing package changes...
(1/4) installing java-runtime-common [#####] 100%
For the complete set of Java binaries to be available in your PATH,
you need to re-login or source /etc/profile
Please note that this package does not support forcing JAVA_HOME as former package java-common did
(2/4) installing java-environment-common [#####] 100%
(3/4) installing jdk-openjdk [#####] 100%
Optional dependencies for jdk-openjdk
  java-rhino: for some JavaScript support
  alsa-lib: for basic sound support [installed]
  gtk3: for the GTK+ 3 look and feel - desktop usage [installed]
(4/4) installing dbeaver [#####] 100%
:: Running post-transaction hooks...
(1/4) Arming ConditionNeedsUpdate...
(2/4) Updating the DBeaver bundles.info database...
(3/4) Updating icon theme caches...
(4/4) Updating the desktop file MIME type cache...

```

VII. Descarga de pgAdmin.

```
yay -S pgadmin4
```

```

yay -S pgadmin4
AUR Explicit (1): pgadmin4-9.14-1
AUR Dependency (4): pgadmin4-desktop-9.14-1, pgadmin4-web-9.14-1, mod_wsgi-6.0.2-1, pgadmin4-server-9.14-1
Sync Make Dependency (5): cmake-4.2-1, postgresql-libs-18.6-1, python-pip-26.2.1-1, syft-1.51.0-1, corepack-0.34.6-1
:: (1/5) Downloaded PKGBUILD: pgadmin4-web
:: (2/5) Downloaded PKGBUILD: pgadmin4-desktop
:: (3/5) Downloaded PKGBUILD: mod_wsgi
:: (4/5) Downloaded PKGBUILD: pgadmin4
:: (5/5) Downloaded PKGBUILD: pgadmin4-server
5 pgadmin4 (Build Files Exist)
4 pgadmin4-desktop (Build Files Exist)
3 pgadmin4-web (Build Files Exist)
2 mod_wsgi (Build Files Exist)
1 pgadmin4-server (Build Files Exist)
=> Packages to clean/build?
=> [N]one [A]ll [Ab]ort [I]nstalled [No]ntInstalled or (1 2 3, 1-3, ~4)
=> n
5 pgadmin4 (Build Files Exist)
4 pgadmin4-desktop (Build Files Exist)
3 pgadmin4-web (Build Files Exist)
2 mod_wsgi (Build Files Exist)
1 pgadmin4-server (Build Files Exist)
=> Diff to show?
=> [N]one [A]ll [Ab]ort [I]nstalled [No]ntInstalled or (1 2 3, 1-3, ~4)
=> n
=> Making package: pgadmin4 9.14-1 (Sat 05 Sep 2026 12:14:51 PM CST)
=> Retrieving sources...
WARNING: Skipping verification of source file PGP signatures.
=> Making package: pgadmin4-desktop 9.14-1 (Sat 05 Sep 2026 12:14:52 PM CST)
=> Retrieving sources...
Downloading pgadmin4-9.14.tar.gz...
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
85 43.51M 85 37.04M 0 0 3.11M 0 0:13 0:11 0:02 2.80M

navega@navega:~$ yay -S pgadmin4
=> Checking buildtime dependencies...
=> Retrieving sources...
=> Removing existing $srcdir/ directory...
=> Extracting sources...
=> Sources are ready:
=> Making package: pgadmin4 9.14-1 (Sat 05 Sep 2026 12:32:18 PM CST)
=> Checking runtime dependencies...
=> Checking buildtime dependencies...
=> WARNING: Using existing $srcdir/ tree
=> Entering fakeroot environment...
=> Tidying install...
=> Removing libtool files...
=> Purging unwanted files...
=> Stripping unneeded symbols from binaries and libraries...
=> Compressing man and info pages...
=> Checking for packaging issues...
=> Creating package "pgadmin4"...
=> Generating .PKGINFO file...
=> Generating .BUILDINFO file...
=> Generating .MTREE file...
=> Compressing package...
=> Leaving fakeroot environment.
=> Finished making: pgadmin4 9.14-1 (Sat 05 Sep 2026 12:32:22 PM CST)
=> Cleaning up...
=> /var/lib/pacman/db.lck is present.
=> There may be another Pacman instance running. Waiting...
loading packages...
resolving dependencies...
looking for conflicting packages...

Packages (1) pgadmin4-9.14-1

:: Proceed with installation? [Y/n] y
(1/1) checking keys in keyring [#####] 100%
(1/1) checking package integrity [#####] 100%
(1/1) loading package files [#####] 100%
(1/1) checking for file conflicts [#####] 100%
(1/1) checking available disk space [#####] 100%
:: Processing package changes...
(1/1) installing pgadmin4 [#####] 100%
checking dependencies...
warning: removing postgresql-libs from target list
:: libvirt optionally requires python-cryptography: for virt-gemu-sev-validate command
:: python optionally requires python-pip: for installing Python packages using tooling that is usually bundled with Python
:: samba optionally requires python-cryptography: lockout_tests.py gp_cert_auto_enroll_ext.py gpo.py and kcrypto.py

Packages (9) cpdpap-1.58.0-3 jsoncpp-1.9.6-3 python-cryptography-50.0.0-1 python-wheel-0.48.0-1 rhash-1.4.6-1 cmake-4.2-1 corepack-0.34.6-1 python-pip-26.2.1-1 syft-1.51.0-1

navega@navega:~$ yay -S pgadmin4
=> Generating .MTREE file...
=> Compressing package...
=> Leaving fakeroot environment.
=> Finished making: pgadmin4 9.14-1 (Sat 05 Sep 2026 12:32:22 PM CST)
=> Cleaning up...
=> /var/lib/pacman/db.lck is present.
=> There may be another Pacman instance running. Waiting...
loading packages...
resolving dependencies...
looking for conflicting packages...

Packages (1) pgadmin4-9.14-1

:: Proceed with installation? [Y/n] y
(1/1) checking keys in keyring [#####] 100%
(1/1) checking package integrity [#####] 100%
(1/1) loading package files [#####] 100%
(1/1) checking for file conflicts [#####] 100%
(1/1) checking available disk space [#####] 100%
:: Processing package changes...
(1/1) installing pgadmin4 [#####] 100%
checking dependencies...
warning: removing postgresql-libs from target list
:: libvirt optionally requires python-cryptography: for virt-gemu-sev-validate command
:: python optionally requires python-pip: for installing Python packages using tooling that is usually bundled with Python
:: samba optionally requires python-cryptography: lockout_tests.py gp_cert_auto_enroll_ext.py gpo.py and kcrypto.py

Packages (9) cpdpap-1.58.0-3 jsoncpp-1.9.6-3 python-cryptography-50.0.0-1 python-wheel-0.48.0-1 rhash-1.4.6-1 cmake-4.2-1 corepack-0.34.6-1 python-pip-26.2.1-1 syft-1.51.0-1

Total Removed Size: 211.84 MiB
:: Do you want to remove these packages? [Y/n]
:: Processing package changes...
(1/9) removing corepack [#####] 100%
(2/9) removing syft [#####] 100%
(3/9) removing python-pip [#####] 100%
(4/9) removing python-wheel [#####] 100%
(5/9) removing python-cryptography [#####] 100%
(6/9) removing cmake [#####] 100%
(7/9) removing rhash [#####] 100%
(8/9) removing jsoncpp [#####] 100%
(9/9) removing cpdpap [#####] 100%
:: Running post-transaction hooks...
(1/4) Updating the MIME type database...
(2/4) Arising ConditionNeedsUpdate...
(3/4) Updating icon theme caches...
(4/4) Updating the desktop file MIME type cache...

```

Comentarios y los problemas a los que te enfrentaste en la instalación.

Me parece que fue adecuado dedicar tiempo al procedimiento de instalación de las herramientas con las que estaremos trabajando para evitar retrasos al momento de comenzar a desarrollar las prácticas. Personalmente, más allá de tener que investigar por separado los comandos requeridos para levantar el servicio y el contenedor en Manjaro, y de esperar por la instalación de pgAdmin (de casi 20 minutos), el proceso me pareció sencillo.